Internet Technologies Lab

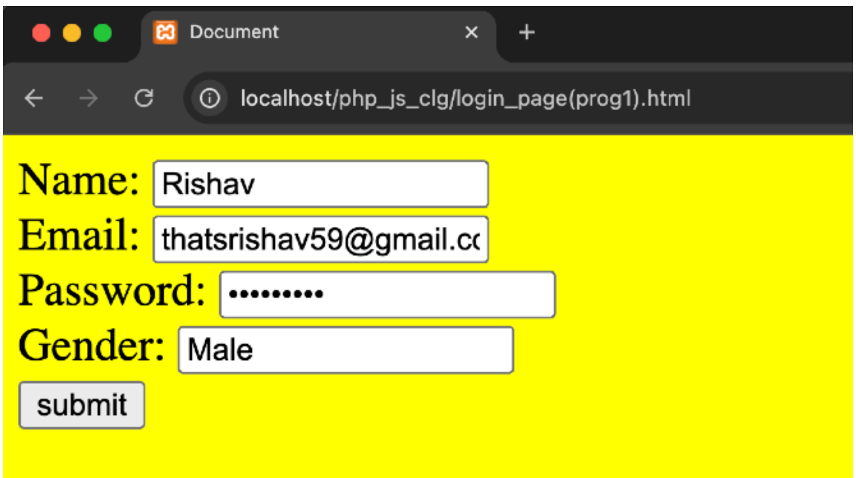
4^{th} semester BCA

Part-A

1) **Design Student Signup Form Using HTML, JavaScript, HTML5 & CSS- (save this file as ".html")**

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Student Signup Form</title>
    <style>
        /* Styling for the form */
        body {
            font-size: 20px;
            background-color: yellow;
            font-family: Arial, sans-serif;
        }
    </style>
    <script>
        function callme() {
            // Getting input values from the form
            var name = document.getElementById("name").value;
            var email = document.getElementById("email").value;
            var password = document.getElementById("password").value;
            var gender = document.querySelector('input[name="gender"]:checked')?.value; // Fetch selected gender

            // Logging values to the console for verification
            console.log("Name: " + name);
            console.log("Email: " + email);
            console.log("Password: " + password);
            console.log("Gender: " + gender);
        }
    </script>
</head>
<body>
    <form>
        <!-- Input fields for user data -->
        Name: <input type="text" name="name" id="name" required> <br>
        Email: <input type="email" name="email" id="email" required> <br>
        Password: <input type="password" name="password" id="password" required> <br>

        <!-- Radio buttons for gender selection -->
        Gender:
        <input type="radio" name="gender" value="Male"> Male
        <input type="radio" name="gender" value="Female"> Female
        <br>

        <!-- Submit button calls the JavaScript function -->
        <input type="button" value="Submit" onclick="callme()">
    </form>
</body>
</html>
```

OUTPUT-



2) Step for installing PHP & Configuring XAMPP-

- The acronym XAMPP stands for cross- Platform, PHP, Apache, MySQL, Perl and PHP for 'PHP hyper text Preprocessor'

- XAMPP is a free and open source web serve that allows you to develop, text and build websites or a local server.

- For PHP development, you need:
  - XAMPP Software
  - Code editor (e.g. Visual Studio Code)

- Download and install XAMPP from the official website - '<u>apachefriends.org</u>'

- When it's done downloading go to your download right click on the setup file (.exe) and select "run as administrator"

- This will take you the setup - XAMPP wizard, click next and you'll be able to select the components you want and then you'll come to the installation folder. You have to select a folder where you want to install XAMPP

- Select the installation language as English, you'll get Bitname for XAMPP

- And you are ready to install while the installation process completes, when it's done click ok.

- Once the installation process is finished, you can now use XAMPP in your project.

- Set the path under System Properties —> enviornment variables —> (location of php) d:\XAMPP\PHP

- To run the PHP program with XAMPP, go to the "XAMPP" folder and open it. You should see a folder named "htdocs". Open it then create a new folder in it.

- Download VS Code Editors from official website: "<u>codevisualstudio.com/download</u>" and download the executable file according to system configuration (32 bit/ 64 bit), follow the simple installation steps.

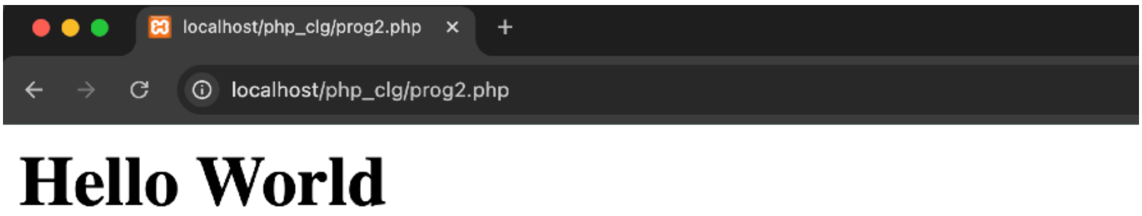
- Next, open your VS code, click on open folder, then go to the location where you saved the folder you created (htdocs/foldername).

- Create a file with the extension “.php” example test.php. PHP is run with the <?php…?> tag.

- After writing the code, open the XAMPP control panel, and start the Apache module by Clicking on start then go to your web browser and in the search bar type local host/Demo (folder name) /test.php, then enter. Output will be displayed - (save this file as “.php”)

```php
<?
    echo "<H1> Hello World </H1>"
?>
```

OUTPUT-



3)Write a program in PHP for type casting of a variable (save this file as ".php")

```php
<?php
$var = 2.23;  // Float variable
echo $var . " " . gettype($var) . "<br>"; // Display value and type

settype($var, "string"); // Convert float to string
echo $var . " " . gettype($var) . "<br>";

settype($var, "integer"); // Convert string to integer
echo $var . " " . gettype($var) . "<br>";

settype($var, "float"); // Convert integer back to float
echo $var . " " . gettype($var) . "<br>";

settype($var, "boolean"); // Convert float to boolean
echo $var . " " . gettype($var) . "<br>";

settype($var, "null"); // Convert variable to NULL
echo gettype($var) . "<br>"; // Display only type since NULL has no output value
?>
```

OUTPUT-

localhost/php_js_clg/set_type(prog3).php

2.23 double
2.23 string
2 integer
2 double
1 boolean
NULL

4) Write a program in PHP to display Multiplication Table using nested for loop (save this file as ".html")

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Multiplication Table</title>
</head>
<body>
   <form action="" method="post">
    Enter Number: <input type="number" name="number" required><br><br>
    <input type="submit" value="Submit">
   </form>

<?php
// Check if form is submitted
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $number = intval($_POST["number"]); // Convert input to an integer

    echo "<h3>Multiplication Table of $number</h3>";

    // Loop to print multiplication table
    for ($i = 1; $i <= 10; $i++) {
        echo "$number x $i = " . ($number * $i) . "<br>";
    }
}
?>
</body>
</html>
```

OUTPUT-

localhost/php_js_clg/table_of_10(prog2).php

Multiplication Table (1 to 10)

1 x 1 = 1	1 x 2 = 2	1 x 3 = 3	1 x 4 = 4	1 x 5 = 5	1 x 6 = 6	1 x 7 = 7	1 x 8 = 8	1 x 9 = 9	1 x 10 = 10
2 x 1 = 2	2 x 2 = 4	2 x 3 = 6	2 x 4 = 8	2 x 5 = 10	2 x 6 = 12	2 x 7 = 14	2 x 8 = 16	2 x 9 = 18	2 x 10 = 20
3 x 1 = 3	3 x 2 = 6	3 x 3 = 9	3 x 4 = 12	3 x 5 = 15	3 x 6 = 18	3 x 7 = 21	3 x 8 = 24	3 x 9 = 27	3 x 10 = 30
4 x 1 = 4	4 x 2 = 8	4 x 3 = 12	4 x 4 = 16	4 x 5 = 20	4 x 6 = 24	4 x 7 = 28	4 x 8 = 32	4 x 9 = 36	4 x 10 = 40
5 x 1 = 5	5 x 2 = 10	5 x 3 = 15	5 x 4 = 20	5 x 5 = 25	5 x 6 = 30	5 x 7 = 35	5 x 8 = 40	5 x 9 = 45	5 x 10 = 50
6 x 1 = 6	6 x 2 = 12	6 x 3 = 18	6 x 4 = 24	6 x 5 = 30	6 x 6 = 36	6 x 7 = 42	6 x 8 = 48	6 x 9 = 54	6 x 10 = 60
7 x 1 = 7	7 x 2 = 14	7 x 3 = 21	7 x 4 = 28	7 x 5 = 35	7 x 6 = 42	7 x 7 = 49	7 x 8 = 56	7 x 9 = 63	7 x 10 = 70
8 x 1 = 8	8 x 2 = 16	8 x 3 = 24	8 x 4 = 32	8 x 5 = 40	8 x 6 = 48	8 x 7 = 56	8 x 8 = 64	8 x 9 = 72	8 x 10 = 80
9 x 1 = 9	9 x 2 = 18	9 x 3 = 27	9 x 4 = 36	9 x 5 = 45	9 x 6 = 54	9 x 7 = 63	9 x 8 = 72	9 x 9 = 81	9 x 10 = 90
10 x 1 = 10	10 x 2 = 20	10 x 3 = 30	10 x 4 = 40	10 x 5 = 50	10 x 6 = 60	10 x 7 = 70	10 x 8 = 80	10 x 9 = 90	10 x 10 = 100

5) Write a program In PHP to Sort an array using Bubble Sort function (save this file ".php")

```php
<h1>Sorting using Bubble Sort Function</h1>
<?php
// Initialize an array
$arr = array(3, 6, 1, 8, 6, 4);
echo "<h4>Before Sorting:</h4>";
print_r($arr); // Print unsorted array

// Bubble Sort Algorithm
for ($i = 0; $i < count($arr); $i++) {
    for ($j = 0; $j < count($arr) - 1 - $i; $j++) { // Optimization: -$i reduces unnecessary comparisons
        if ($arr[$j] > $arr[$j + 1]) {
            // Swap elements if they are in the wrong order
            $temp = $arr[$j];
            $arr[$j] = $arr[$j + 1];
            $arr[$j + 1] = $temp;
        }
    }
}

echo "<h4>After Sorting:</h4>";
print_r($arr); // Print sorted array
?>
```

OUTPUT-

localhost/php_js_clg/bubble_sort(pro5).php

Original Array: Array ([0] => 64 [1] => 32 [2] => 25 [3] => 12 [4] => 22 [5] => 11 [6] => 90)
Sorted Array: Array ([0] => 11 [1] => 12 [2] => 22 [3] => 25 [4] => 32 [5] => 64 [6] => 90)

6) Design the personal information form, submit and retrieve the form data using php $_POST,$_GET,$ REQUEST variable (save this file as ".php"

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
</head>
<body>
   <!-- Form to collect personal information -->
   <!-- The form submits data using POST method -->
   <!-- 'course' and 'college' are passed as GET parameters in the URL -->
   <form action="index.php?course=BCA&college=Presidency" method="post">

    <!-- Input field for First Name -->
    First Name: <input type="text" name="fname"><br><br>

    <!-- Input field for Last Name -->
    Last Name: <input type="text" name="lname"><br><br>

    <!-- Input field for Email ID -->
    Email id: <input type="email" name="email"><br><br>

    <!-- Radio buttons for selecting Gender -->
    Select Gender:
    <input type="radio" id="" name="gender" value="female"> Female
    <input type="radio" id="" name="gender" value="male"> Male
    <br><br>

    <!-- Input field for Age -->
    Age: <input type="number" name="age"><br><br>

    <!-- Input field for Address -->
    Address: <input type="text" name="address"><br><br>

    <!-- Submit button to send form data -->
    <input type="submit" value="submit">
</form>

<?php
// Retrieving form data using the POST method
$fname = $_POST["fname"]; // First Name
$lname = $_POST["lname"]; // Last Name
$email = $_POST["email"]; // Email ID
$gender = $_POST["gender"]; // Gender

// Retrieving Age and Address using REQUEST (works for both GET & POST)
$age = $_REQUEST["age"];
$address = $_REQUEST["address"];

// Retrieving Course and College using GET method
$course = $_GET["course"];
$college = $_GET["college"];

// Displaying the submitted data
echo "<br>Your First Name: ".$fname."<br>";
echo "Your Last Name: ".$fname."<br>"; // Might be a mistake, should this be $lname instead of $fname?
echo "Your Email: ".$email."<br>";
echo "Your gender: ".$gender."<br>";
echo "Your age: ".$age."<br>";
echo "Your address: ".$address."<br>";
echo "Your course: ".$course."<br>";
echo "Your college: ".$college."<br>";
?>

</body>
</html>
```

OUTPUT-

After clicking Submit-

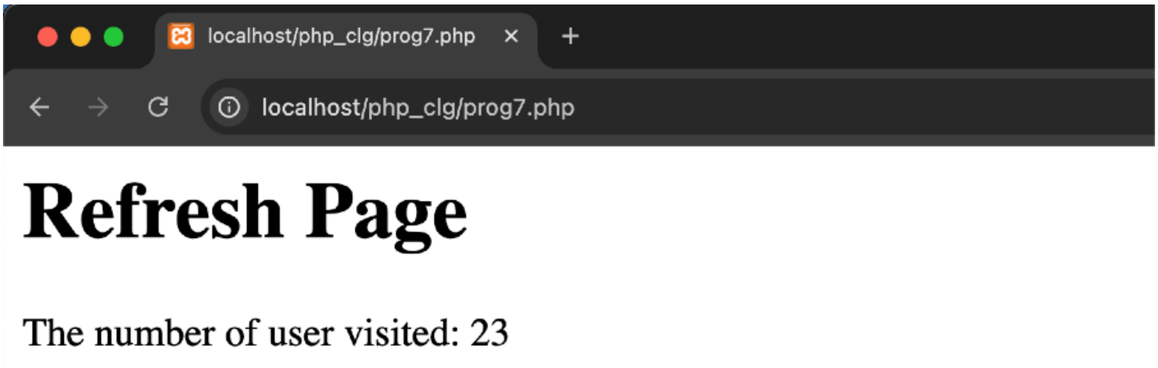
Your First Name: Sahid
Your Last Name: Sahid
Your Email: sahidakhtar521@gmail.com
Your gender: male
Your age: 20
Your address: Rachi
Your course: BCA
Your college: Presidency

7) Write a PHP program to keep track of the number of visitors visiting the web page and to display this count of visitors, with proper headings. (Save this file as ".php")

Note: (Create a text file with"count.txt" name and set the value to "1")

```php
<?php
echo "<h1>Refresh Page</h1>";
$file = "count.txt";

// Check if file exists, if not, create it with an initial count of 1
if (!file_exists($file)) {
    file_put_contents($file, "1");
} else {
    // Read current count and increment
    $c = file_get_contents($file);
    file_put_contents($file, $c + 1);
    echo "Number of visitors: " . ($c + 1);
}
?>
```

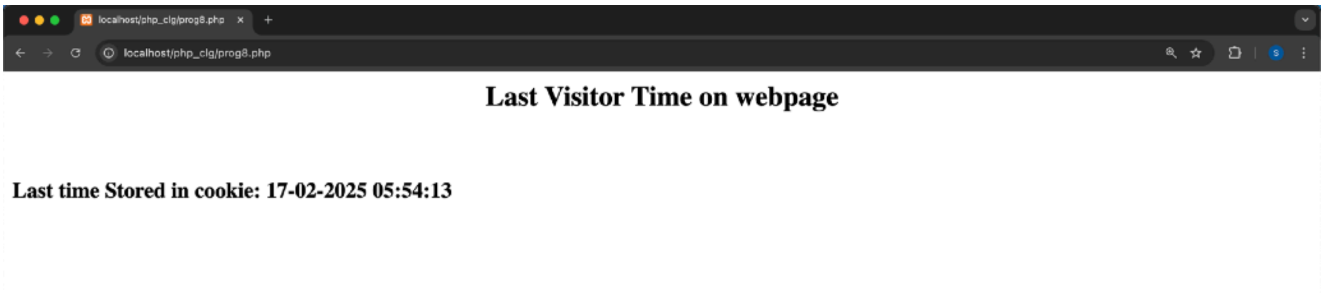
OUTPUT-



8) Write a PHP program to use a cookie to remember the day of the last login from a user and display it when run. (Save this file as ".php")

```
<html>
<body>
    <center><h2>Last Visitor Time</h2></center>
    <br>
<?php
// Check if cookie exists to display last visit time
if (isset($_COOKIE["time"])) {
    $time = $_COOKIE["time"];
    echo "<h3>Last Visit: $time</h3>";
} else {
    echo "<h3>First Time Visit</h3>";
}

// Store current visit time in a cookie (expires in 1 year)
$time = date("d-m-Y H:i:s");
setcookie("time", $time, time() + (365 * 24 * 60 * 60)); // 1-year expiration
?>
</body>
</html>
```

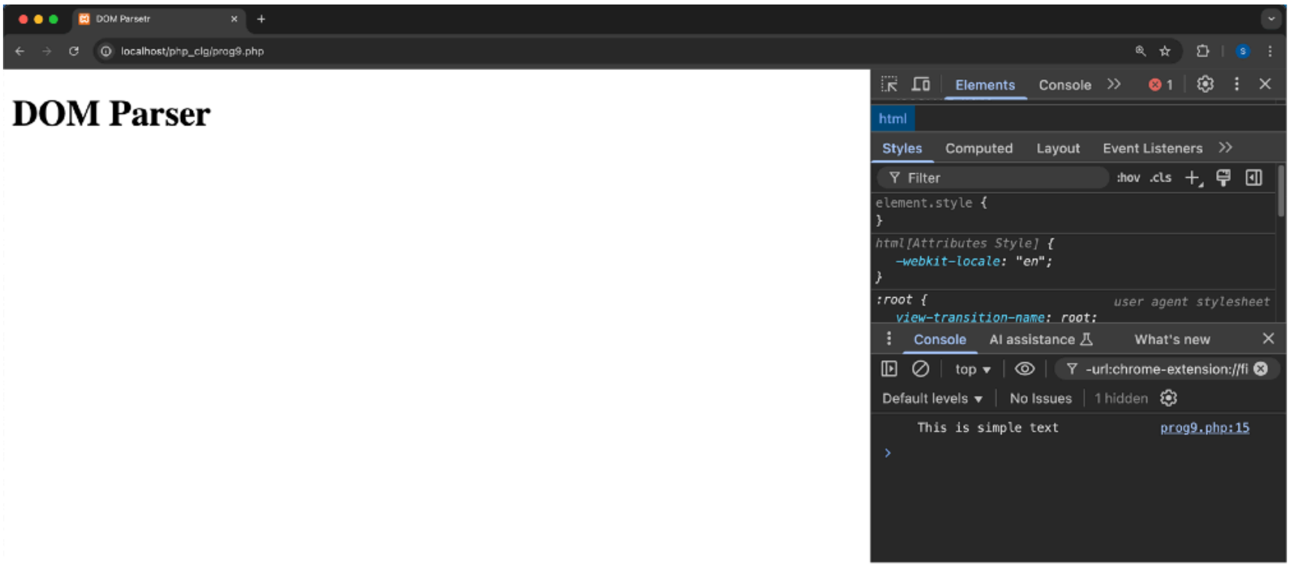
OUTPUT-



9) Write a program to demonstrate DOM parser (save this file has “.html”)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>DOM Parser</title>
</head>
<body>
    <h1>DOM Parser Example</h1>
    <script>
        // Creating a DOMParser instance
        const parser = new DOMParser();

        // Parsing a string as an HTML document
        const parserDocument = parser.parseFromString("<body><h1 id='title'>This is h1 tag</h1></body>", "text/html");

        // Select the h1 element inside the parsed document
        const title = parserDocument.getElementById("title");

        // Modify the text content of the selected element
        title.textContent = "Modified Text";

        // Log the modified document to the console
        console.log(parserDocument.body.innerHTML);
    </script>
</body>
</html>
```

OUTPUT-



10) Create a registration form which stores the registration details into the database table.

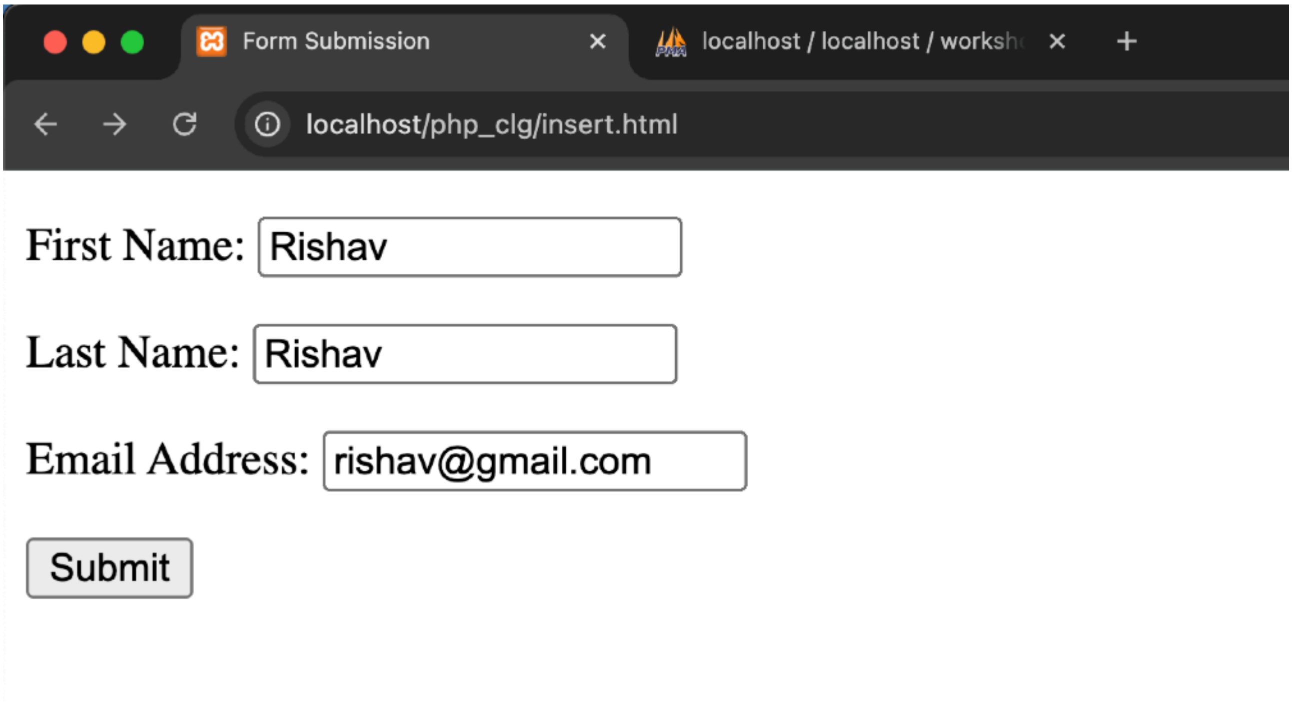
PHP code- (save this file as store ".php")

```php
<?php
// Database connection
$conn = mysqli_connect("localhost", "root", "", "workshop");

// Check connection
if (!$conn) {
    die("ERROR: Could not connect. " . mysqli_connect_error());
}

// Check if form is submitted via POST
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    // Sanitize input data
    $first_name = mysqli_real_escape_string($conn, $_POST['first_name']);
    $last_name = mysqli_real_escape_string($conn, $_POST['last_name']);
    $email = mysqli_real_escape_string($conn, $_POST['email']);

    // Insert data into database
    $sql = "INSERT INTO persons (first_name, last_name, email) VALUES ('$first_name', '$last_name', '$email')";

    if (mysqli_query($conn, $sql)) {
        echo "Records added successfully.";
    } else {
        echo "ERROR: Could not execute $sql. " . mysqli_error($conn);
    }
}

// Close the database connection
mysqli_close($conn);
?>
```

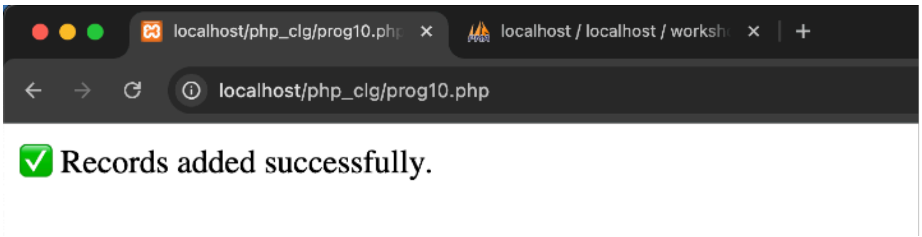
HTML Code- First run this program ("insert.php") this will call another program. (Save this file as "html")

```html
<!DOCTYPE html>
<html>
<head>
    <title>Registration Form</title>
</head>

<body>
    <!-- Form to collect user details -->
    <!-- The form submits data to 'store.php' for processing -->
    <form action="store.php">

        <!-- Input field for First Name -->
        <p>
            <label for="firstName">First Name:</label>
            <input type="text" name="first_name" id="firstName">
        </p>

        <!-- Input field for Last Name -->
        <p>
            <label for="lastName">Last Name:</label>
            <input type="text" name="last_name" id="lastName">
        </p>

        <!-- Input field for Email Address -->
        <p>
            <label for="emailAddress">Email Address:</label>
            <input type="text" name="email" id="emailAddress">
        </p>

        <!-- Submit button to send form data -->
        <input type="submit" value="Submit">

    </form>

</body>
</html>
```

OUTPUT-



After entering the details and clicking the Submit button this output will come-



After this open “phpmyadmin” site and open your database and you will see this output-